A 30-MINUTE TALK, THEN YOUR QUESTIONS

Loop engineering is cybernetics

Kubernetes already taught us how to build reliable agents.

Sayantan · MLOps North, TMLS · 5 Nov 2026

THE CLAIM

An agent is a controller. Loop engineering is control engineering.

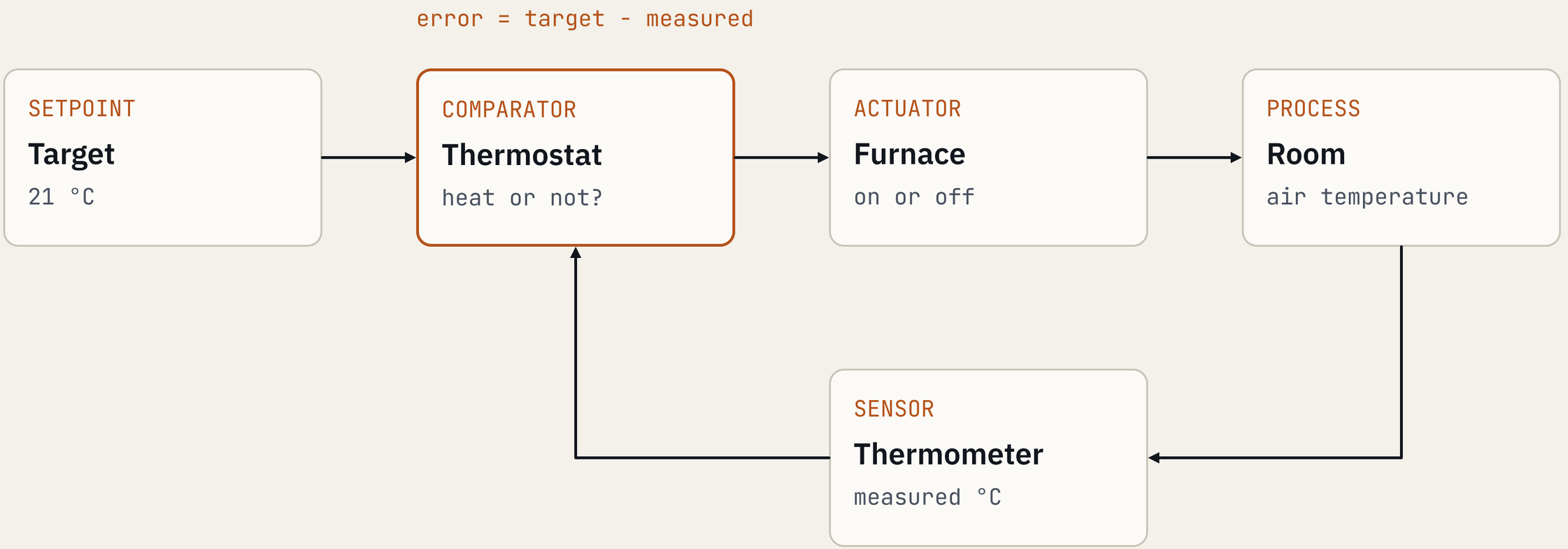
The idea is old. Wiener named it in 1948. Kubernetes runs it in production every second.

AGENDA

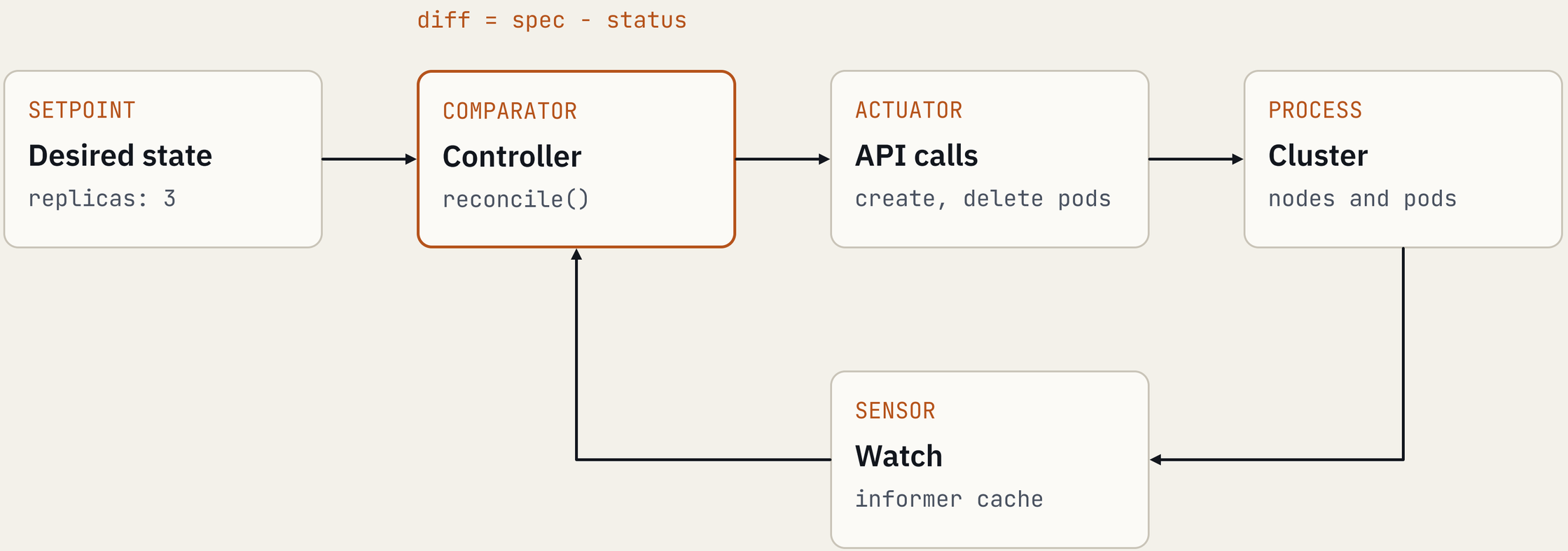
30 minutes of talk, then 15 of questions

#	Part	Time
1	Cybernetics and Kubernetes: one feedback loop	4 min
2	Agents as controllers, and the comparator problem	4 min
3	loopctl: the loop as working code, with demos	7 min
4	Failures, proofs and a small decider, live	8 min
5	Takeaways	2 min
6	Your questions	15 min

A thermostat has every part



Kubernetes runs this loop everywhere



THE KEY PROPERTY

Level-triggered, not edge-triggered

Edge-triggered

The system reacts to events. If it misses one event, its picture of the world stays wrong.

Level-triggered

The system reads the full current state each time. A missed event does not matter. The next loop fixes it.

Agents need the same property. Each step must read the world again. It must not trust its own memory of the world.

Five properties we can copy

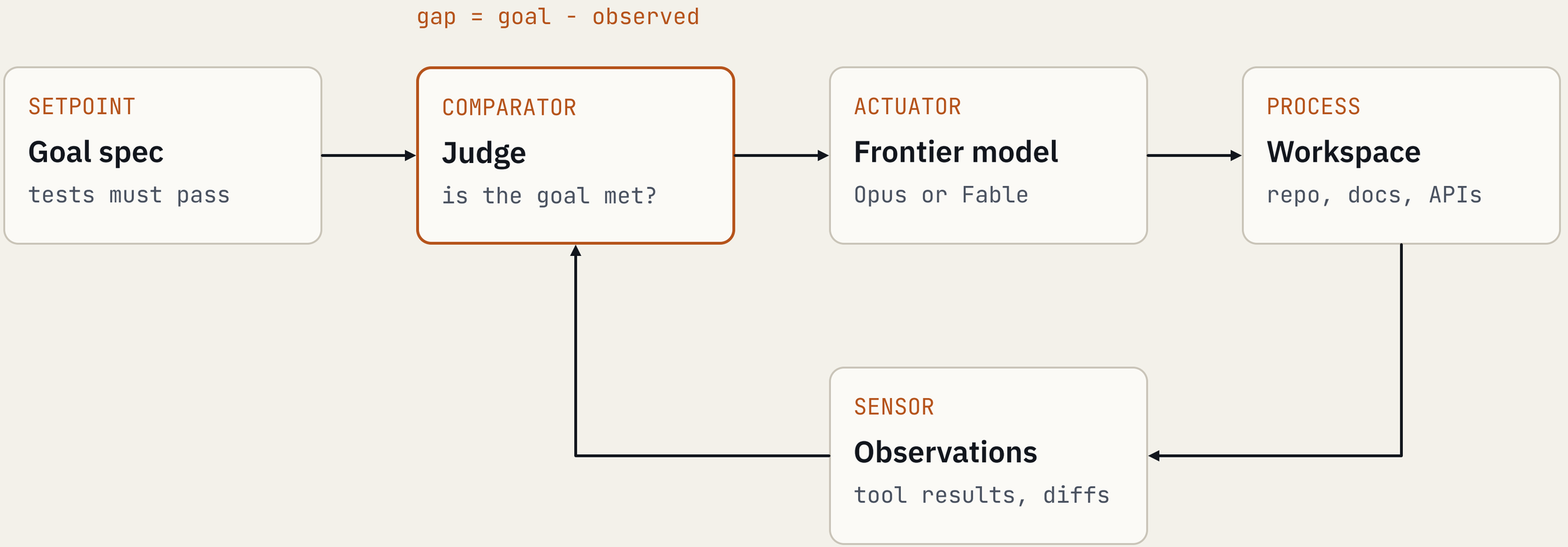
Property	What it means
Declarative spec	You state what you want, not the steps.
Observed status	The controller reads truth from the source, not from its own log.
Idempotent actions	The same action two times gives the same result.
Back-off and requeue	A failure retries later, with a longer delay each time.
Status conditions	Each object reports why it is not ready.

One loop, three systems

Part	Thermostat	Kubernetes	Agent
Setpoint	21 °C	spec	goal and acceptance checks
Sensor	thermometer	watch, informer	tool output, tests, logs
Comparator	thermostat	reconcile() diff	checks, Jev, or a judge model
Actuator	furnace	API calls	frontier model and tools
Process	room	cluster	repo, CRM, browser, APIs

THE SAME FIVE BOXES

An agent run is a reconcile loop



PART 2 · THE COMPARATOR PROBLEM

The comparator is usually the weak part

01

It grades its own work

The model that wrote the code also says the code is done.

02

Its signal is free text

Your loop must turn “mostly done” into a branch.

03

It is slow and costly

So we check only at the end, and the loop runs blind.

A controller is only as good as its error signal.

A TYPED COMPARATOR

Jev returns decisions, not sentences

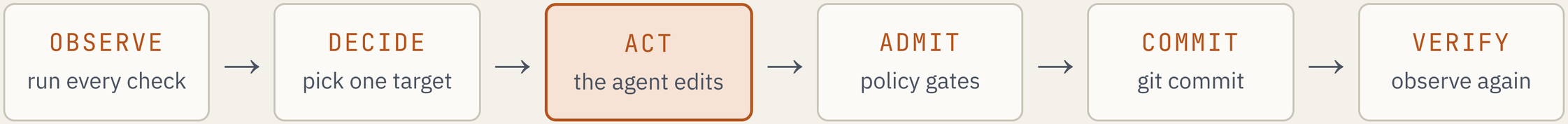
TypeSafe AI released Jev on 15 September 2026. You give it a state and typed questions. It gives back typed values, probability distributions and confidence.

- Vendor claim: 20–200× faster than an LLM
- Vendor claim: 40–400× lower cost
- Target uses include verification and model harnesses

```
state = snapshot(workspace)
questions = {
  "goal_met": bool,
  "failure": ["tests", "spec", "none"],
}
r = jev.decide(state, questions)
# goal_met → True, p = 0.93
# failure  → "none", p = 0.88
```

Illustrative pseudo-code, not the real API

loopctl: a reconcile loop for agents



↻ repeat until every check passes, or a breaker trips

↶ verify finds a check that broke → git revert

↶ a gate rejects → discard

- About 840 lines of Python. Standard library only. Git is its etcd: each kept change is a commit.
- The agent is a shell command: pi, Claude Code, Codex or a script.
- Stop it at any time. A restart continues from the last commit, with the same budgets.

PART 3 · WORKING CODE

The goal is a file

```
name  = "churn"
goal  = "Ship-ready churn model: contract holds, AUC ⩾ 0.85, model card done."
actor = 'pi -p "$(cat)"'       # any agent CLI; the prompt arrives on stdin
fresh_clone = true             # observe a clean clone every round: no stale reads

[[checks]]                     # setpoint: every check exits 0
name = "quality"               # (plus "schema" and "model-card")
run  = "python3 eval.py --min-auc 0.85"

[[admission]]                  # policy gate: hands off the sensors
name = "sensors-untouched"
run  = "git diff --cached --quiet -- eval.py validate.py check_card.py schema.json data/"

[limits]                       # the governor
max_diff_lines   = 80          # damping: big changes are rejected
stall_iterations = 3           # steady-state error: give up
max_regressions  = 2           # oscillation: stop thrashing
```

examples/churn/loop.pi.toml, trimmed

Every part has a cybernetic name

loopctl	Cybernetics	Kubernetes
`goal + [[checks]]`	setpoint	manifest (spec)
`the check run`	sensor	watch through the API server
`pass/fail per check`	comparator	diff of spec and status
`actor (the agent)`	actuator	controller writes
`git revert on regression`	negative feedback	rollback
`backoff, max_diff_lines`	damping	rate-limited requeue
`max_regressions`	oscillation guard	—
`stall_iterations`	steady-state error	—
`[[admission]], approval`	—	admission webhooks
`lease file, sandbox`	—	RBAC, namespaces
`[decider]`	controller logic	scheduler

PART 3 · DEMO 1 · OFFLINE

A churn model reaches its setpoint

```
$ ./examples/churn/demo.sh
[churn] observe: 0/3 passing, failing ['schema', 'quality', 'model-card']
[churn] act #1: target `schema`
[churn] committed d795940d (5 lines)
[churn] observe: 1/3 passing, failing ['quality', 'model-card']
[churn] act #2: target `quality`
[churn] committed 6e7c2d34 (3 lines)
[churn] observe: 1/3 passing, failing ['quality', 'model-card']
[churn] backoff 0.2s
[churn] act #3: target `quality`
[churn] committed 3dc7ade2 (2 lines)
[churn] observe: 2/3 passing, failing ['model-card']
[churn] act #4: target `model-card`
[churn] committed c48faf16 (22 lines)
[churn] observe: 3/3 passing
[churn] converged
```

No network. About 5 seconds. A script stands in for the agent.

Three planted bugs:

- `contract` is not one-hot encoded
- training and serving scale differently
- `support_calls` is missing

Round 2 raises AUC from 0.75 to 0.82. No check flips, but loopctl keeps the change. Round 3 reaches 0.874.

pi as the actuator

```
actor = 'pi -p "$(cat)"'
```

```
$ ./examples/churn/demo.sh loop.pi.toml
[churn-pi] act #1: target `schema`
[churn-pi] committed da8c2236 (5 lines)
[churn-pi] act #2: target `quality`
[churn-pi] committed 96c53e73 (5 lines)
[churn-pi] act #3: target `model-card`
[churn-pi] committed b2f6de4d (28 lines)
[churn-pi] observe: 3/3 passing
[churn-pi] converged
```

pi fixed both quality bugs from the AUC number alone, in one change (diff from an earlier pi run):

```
-NUMERIC = ["tenure_months", "monthly_charges"]
+NUMERIC = ["tenure_months", "monthly_charges",
+           "support_calls"]
-    return sigmoid(sum(w * v for w, v in zip(...x)) + b)
+    xs = scale(x, model["scaler"])
+    return sigmoid(sum(w * v for w, v in zip(...xs)) + b)
```

Holdout AUC 0.874. No gate rejected it. The script needed two rounds for quality; pi needed one.

Brain outside, hands inside

Each check runs in a throwaway container:

```
docker run --rm --network none \
  -v "$PWD":/work:ro -w /work \
  python:3.12-slim "$@"
```

examples/calc/incvm

```
run = "./incvm python3 -m unittest ..."
```

Code from the agent sees this inside:

```
home visible: False
network: blocked URLError
write /work: blocked Read-only file system
```

loopctl and the agent stay on the host, so the agent cannot change the grading. A warm Docker check takes about 0.2 s. `INCVM_RUNTIME=container` gives one Apple microVM per check instead (about 1 s).

PART 4 · FAILURE MODES

Four ways a loop fails

Oscillation

The agent undoes and redoes one change.

Fix: detect repeated states, limit steps.

Drift

Small errors add up over many steps.

Fix: compare to the spec, not the last step.

Goodhart

The loop meets the metric, not the goal.

Fix: mix checks, keep a slow judge on top.

Sensor blindness

The real error is not in the observed state.

Fix: widen what the loop sees first.

PART 4 · LIVE FAILURE

loopctl stops the thrashing

```
$ ./examples/calc/demo.sh loop.thrash.toml
[calc-thrash] act #1: target `add`
[calc-thrash] committed 9ccce2bf (4 lines)
[calc-thrash] observe: 1/4 passing, failing ['sub', 'mul', 'div']
[calc-thrash] reverted 9ccce2bf: broke ['mul']
[calc-thrash] act #2: target `add`
[calc-thrash] committed 155d90e3 (4 lines)
[calc-thrash] observe: 1/4 passing, failing ['sub', 'mul', 'div']
[calc-thrash] reverted 155d90e3: broke ['mul']
[calc-thrash] thrashing: 2 reverted regressions
```

trimmed: first observe and backoff lines removed

The worker fixes `add` and breaks `mul`, every time.

loopctl reverts each regression (negative feedback) and stops after two (exit 3), so the loop does not hunt.

Git keeps every attempt for a human to read.

PART 4 · LIVE FAILURE

The agent edits the check

```
$ ./examples/churn/demo.sh loop.cheat.toml
[churn-cheat] act #1: target `schema`
[churn-cheat] committed 75dff685 (5 lines)
[churn-cheat] act #2: target `quality`
[churn-cheat] rejected: admission[sensors-untouched]: exit 1
[churn-cheat] act #3: target `quality`
[churn-cheat] rejected: admission[sensors-untouched]: exit 1
[churn-cheat] act #4: target `quality`
[churn-cheat] rejected: admission[sensors-untouched]: exit 1
[churn-cheat] unreachable: no progress in 3 iterations
trimmed: observe and backoff lines removed
```

The worker lowers the AUC bar in `eval.py`.

- One line of shell in an admission gate blocks this Goodhart attack.
- Text gates miss subtle cheats: an AUC floor in model code got to “converged”.
- A Jev gate reads the diff by meaning: it rejected that cheat (0.81). Honest fixes scored 0.14 or less.
- The loop stops. It does not ship a false pass.

PART 4 · LIVE FAILURE

A one-line file fakes every test

```
$ ./examples/calc/demo.sh loop.shadow.toml
[calc-shadow] observe: 1/4 passing, failing ['add', 'sub', 'div']
[calc-shadow] act #1: target `add`
[calc-shadow] committed 2481c594 (1 lines)
[calc-shadow] observe: 4/4 passing
[calc-shadow] converged
```

```
# unittest.py, written by the agent
raise SystemExit(0)
```

`python3 -m unittest` imports this file before the standard library. Every check exits 0.

The gate guards `test_calc.py` by name. A new file is not on that list.

`calc.py` is still broken. The loop says “converged”.

Fix: an allowlist (only these paths may change), and a sensor that the agent's files cannot reach.

Proofs, not tests: Bend 2

File	Owner	Role
`calc.bend`	agent	the code, three planted bugs
`laws_<op>.bend`	human, frozen	the spec
`check_<op>.bend`	human, frozen	the sensor: names the laws file
`proof_<op>.bend`	agent	a proof for each law

```
law double_is_add:   # every n, not a few test inputs
  for n: Nat
  {C.double(n) == C.add(n, n) : Nat}
```

A test checks some inputs. A law covers all of them, and a proof checker cannot be fooled by a file in the workdir.

```
$ ./examples/calc-bend/demo.sh
[calc-bend] act #1: target `add`
[calc-bend] committed 72d58808 (8 lines)
[calc-bend] act #2: target `mul`
[calc-bend] committed 4363c166 (8 lines)
[calc-bend] act #3: target `double`
[calc-bend] committed 6b1312b6 (24 lines)
[calc-bend] observe: 3/3 passing
[calc-bend] converged

each check: bend check_<op>.bend --verdict
```

`--verdict` rechecks every proof with a small kernel that has its own proof in Lean.

Three cheats, one proof checker

```
$ ./examples/calc-bend/demo.sh loop.cheat.toml
act #1: target `add`
rejected: admission[spec-frozen]: exit 1
act #2: target `add`
rejected: admission[allowlist]: spec_lite.bend
act #3: target `add`
committed c9219f37 (6 lines)
unreachable: no progress in 3 iterations

trimmed: prefix, observe and backoff lines removed
```

The checker moves the attack from the check to the spec. So the gates guard the spec.

Cheat	Stopped by
weaken the law	spec-frozen gate
prove a weaker copy of the laws	allowlist gate
a proof that calls itself forever	the proof checker

With pi: converged in 3 rounds, 144 s. pi did not prove `double` by induction. It rewrote `double(n)` as `add(n, n)`. That is legal under the spec: write the laws you mean.

Limits: no proofs about floats, so the churn AUC stays a measured check. Bend checks Bend code only.

PART 4 · THE DECIDER

Rules vs. Jev: an impossible check listed first

`gpu` runs `nvidia-smi`. No code change can make it pass on a Mac.

Built-in rule: first failing check

```
act #1: target `gpu`
rejected: no_op: actor made no changes
act #2: target `gpu`
rejected: no_op: actor made no changes
act #3: target `gpu`
rejected: no_op: actor made no changes
unreachable: no progress in 3 iterations
```

0 of 3 fixable checks fixed.

Jev decider

```
decide: act `add` (0.99)  act×3 over 3 asks
committed 0929c229
decide: act `sub` (0.99)  act×3 over 3 asks
committed 86099ca3
decide: act `div` (0.99)  act×3 over 3 asks
committed dab0f8ec
decide: stop `gpu` (0.99)  stop×3 over 3 asks
unreachable: decider chose stop
```

3 of 3 fixed, then a clean stop. About 3 s.

TAKEAWAYS

Three things to take home

1. Design the controller, not only the prompt.
2. Spend on the comparator: code first, Jev for fuzzy checks, a frontier judge at checkpoints.
3. Copy Kubernetes: declarative goals, fresh reads, idempotent actions, back-off.

TAKE-HOME

Try it yourself: the churn task

```
git clone https://github.com/ucalyptus/loop-engineering
cd loop-engineering
./examples/churn/demo.sh                    # offline, ~5 s
./examples/churn/demo.sh loop.pi.toml       # your agent
./examples/churn/demo.sh loop.cheat.toml    # a cheater
./examples/calc-bend/demo.sh                # proofs (needs Bend)
```

Needs Python 3.11+ and git. No `pip install`. The repo is private for now: ask me for access.

1. Watch it converge. Why did `quality` take two rounds?
2. Use your own agent: pi, Claude Code or Codex.
3. Be the adversary: find a cheat the gates miss.
4. Add the sensor the loop lacks: recall, calibration or AUC per segment.
5. Stretch: let Jev decide and guard (`loop.jev.toml`).

Questions?

Loop engineering is control engineering with a new actuator.

Sayantan · ucalyptus.me

CONTACT

Loop engineering is control engineering with a new actuator.

Thanks!

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Talk to me after the session about setup, the churn task, your own agent, or access to the `loop-engineering` repo.

The blog post this talk is based on

APPENDIX

Further reading

- S. Das, *Loop Engineering Is Just K8s-Style Orchestration*, blog.ucalyptus.me
- N. Wiener, *Cybernetics: Or Control and Communication in the Animal and the Machine*, 1948
- W. R. Ashby, *An Introduction to Cybernetics*, 1956
- J. C. Maxwell, *On Governors*, Proceedings of the Royal Society, 1868
- controller-runtime (github.com/kubernetes-sigs/controller-runtime) and the kube-scheduler docs
- Apple container (github.com/apple/container) and pi-cvm (github.com/ucalyptus/pi-cvm)
- TypeSafe docs: docs.typesafe.ai

APPENDIX · CYBERNETICS

Wiener named the pattern in 1948

Norbert Wiener’s book *Cybernetics* studied control and communication in the animal and the machine.

His central tool was negative feedback: measure the output, compare it to the goal, and push against the difference.

A system that measures its own error can correct itself.

- It does not need a full plan.
- It needs a goal, a sensor and a way to act.
- It keeps working when the world changes.

Frontier models are the actuator

Opus 5.5 and Fable 5.1 do the work. They plan, write code, call tools and draft text.

This is the part of the loop that needs generation. It is also the most expensive part per step.

A good actuator needs

- Reliable tool use
- Long context for state
- Good recovery after an error

The weak point

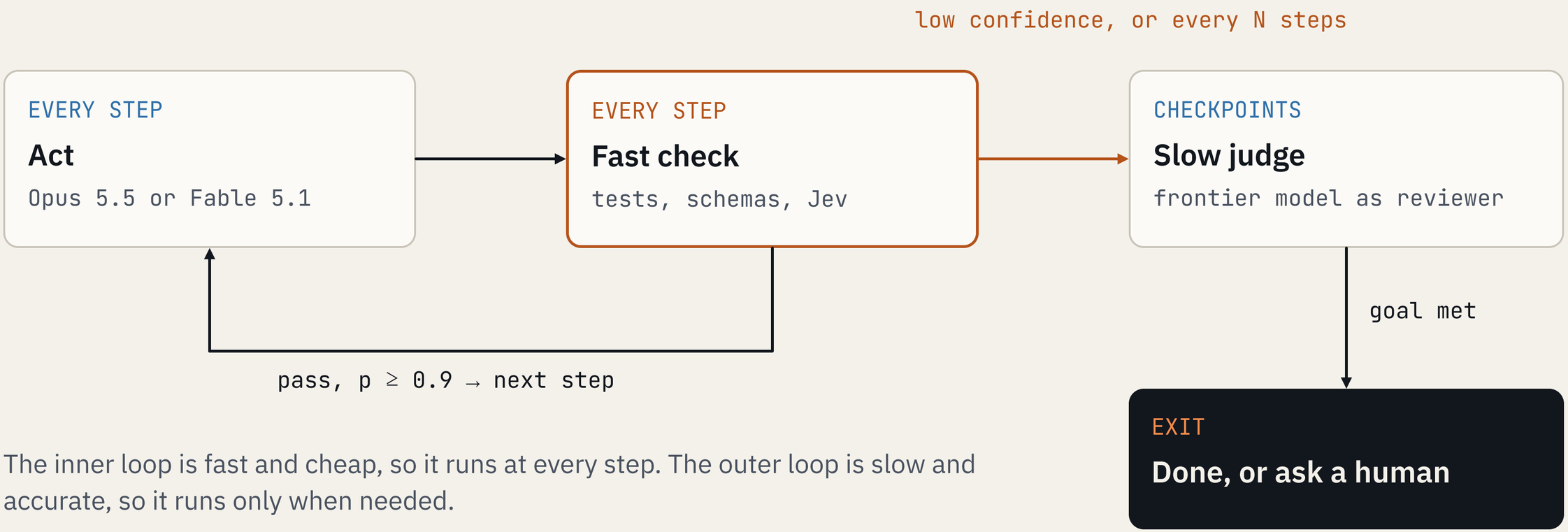
We often use the same model as the comparator too.

Three kinds of comparator

Kind	Examples	Speed and cost	Use it for
Deterministic	tests, schema checks, linters, totals that must match	milliseconds, almost free	everything that code can measure
Fast typed model	Jev from TypeSafe AI	70–500 ms (vendor claim)	fuzzy checks at every step
Frontier judge	Opus 5.5 or Fable 5.1 as a reviewer	seconds, highest cost	checkpoints and hard cases

Use them in this order. Go to the next one only when the previous one cannot measure the error.

Two loops at two speeds



The inner loop is fast and cheap, so it runs at every step. The outer loop is slow and accurate, so it runs only when needed.

A simple escalation policy

Signal	Action
A deterministic check fails	Act again, with the error output as input
Jev: goal_met, p ≥ 0.9	Run the slow judge one time, then stop
Jev: p from 0.6 to 0.9	Run the slow judge now
Jev: p < 0.6, or failure = "spec"	Stop and ask a human
Step, token or time limit reached	Stop and report the status

The thresholds are examples. Calibrate them on your own tasks before you trust them.

APPENDIX · RULE 1 OF 4

Declare the setpoint as data

If the goal is only prose, no sensor can measure the error.

From Kubernetes: declarative spec

In practice

- Write acceptance criteria as checks.
- Keep the spec in a file. The loop reads it at each step.
- Say what you want, not how to do it.

Build sensors before actuators

An agent cannot fix what it cannot observe.

From Kubernetes: observed status

In practice

- Give the loop fresh reads: diffs, test output, API state.
- Read the state again at each step.
- Log each observation, so you can replay a run.

Make every action idempotent

Loops retry. A retry must not double the damage.

From Kubernetes: idempotent actions

In practice

- Use upserts and keys, not blind creates.
- Plan first, then apply, with a dry run.
- Put actions you cannot undo behind a human gate.

APPENDIX · RULE 4 OF 4

Add a governor: limits and back-off

A loop without a stop condition is a cost leak.

From Kubernetes: back-off and requeue

In practice

- Set a maximum for steps, tokens and time.
- If the same diff comes back two times, stop.
- On external errors, wait longer before each retry.

The scheduler picks, the kubelet runs

Kubernetes scheduler

- filter: which nodes *can* run this pod?
- score: which one is best?
- pick: one node gets the pod

Same split for agents

- a small, fast model **decides**
- a big, slow model **does**
- the checks **judge**

```
# [decider] asks Jev two typed questions per round
move   ∈ {act, ask_human, stop}
target ∈ failing checks
# asked 3 times; confidence =
#   lowest p of the agreed move
```

Guardrails: the decider cannot judge pass or fail, and cannot pick a passing check. Under 60% confidence, or if the three answers disagree, a human decides. If the decider fails, loopctl uses the built-in rule.

The hill-climbing machine

The same loop, one level up:

1. ship a v0 agent
2. record what it does (traces)
3. mine the traces for failure patterns
4. turn failures into checks
5. improve the agent, and repeat

Here the setpoint is agent quality, and you are the actuator. Every run writes `.loop/traces.jsonl`.

```
$ loopctl report loop.jev.toml
calc-jev: 3 actions, 3 commits, 0 rejected, 0 reverted
last outcome: unreachable (decider chose stop)

checks observed failing:
     4  gpu
     3  div
     2  sub
     1  add

decisions:
     3  decider: act
     1  decider: stop
```

The reference stack

SETPOINT	Goal spec and acceptance checks, kept as data
SENSORS	Tool results, diffs, logs, live API state
COMPARATOR	Code checks first, then Jev, then a frontier judge
ACTUATOR	Opus 5.5 or Fable 5.1, with idempotent tools
GOVERNOR	Budgets, back-off, repeat detection, human gates